

Class 1 — Lesson Notes

Prompt Engineering: Talking to AI the Right Way

Big Idea

Prompt Engineering is the skill of asking the right questions in the right way so that AI gives you the best possible answers.

1. What Is an LLM?

LLM stands for Large Language Model. ChatGPT, Gemini, and Claude are all LLMs. Before you can write good prompts, you need to understand what these tools actually are — and what they are not.

What an LLM does

- Reads an enormous amount of text from the internet during training
- Learns patterns in how words and sentences are arranged
- When you type something, it predicts the most likely next word — one word at a time

Example

You type: "I love eating ice ____"

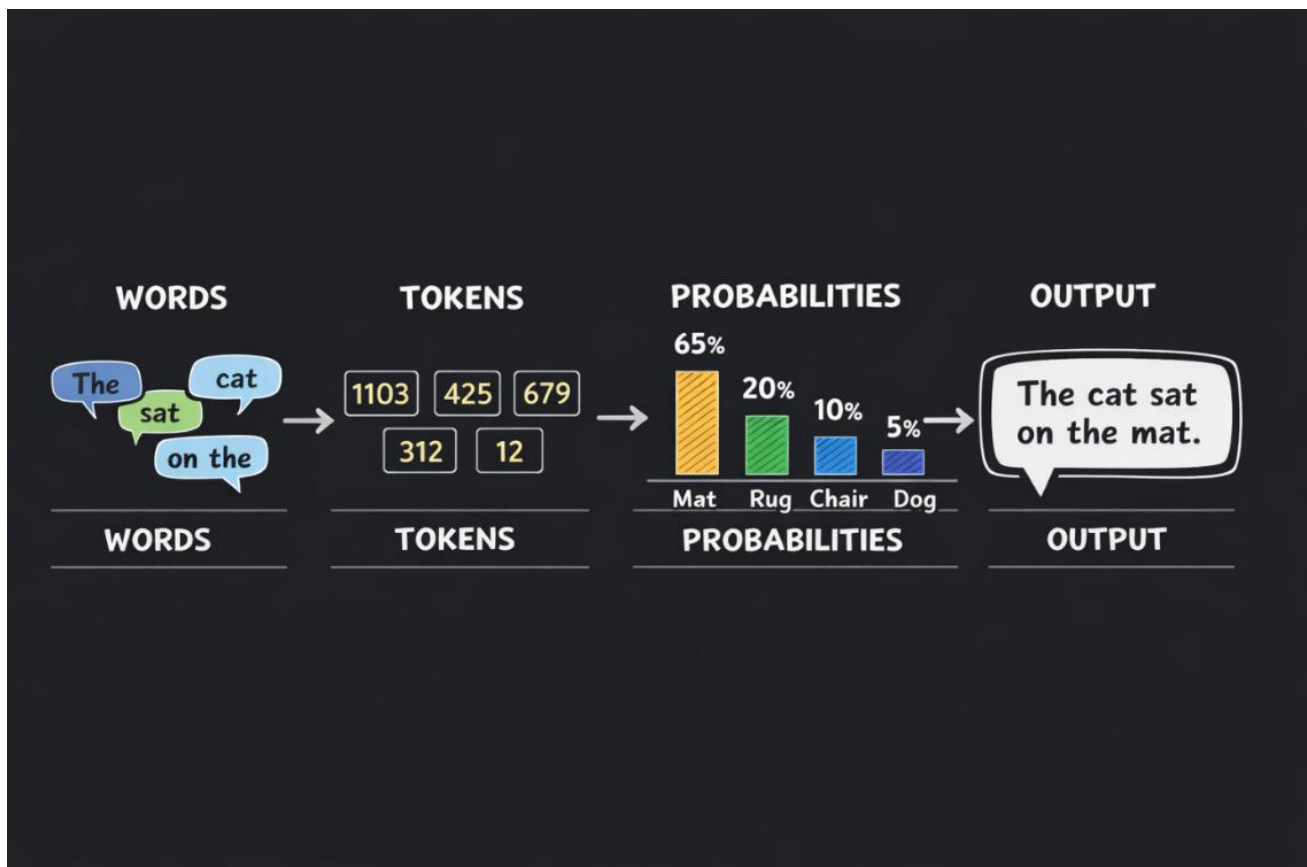
The LLM predicts: cream — because that word follows "ice" most often in its training data.

Key takeaway

An LLM is a text-prediction machine, not a thinking machine. It recognises patterns and predicts what comes next — it does not understand your words the way a human does.

2. Tokens — How AI Actually Reads Text

An LLM does not read words the way we do. It first breaks text into smaller units called tokens, converts them to numbers, and works with those numbers. The diagram below shows how this works in practice.



Key things to notice from the diagram:

- Words are split into tokens — some words become multiple tokens
- Each token is converted to a number the model can process
- The model assigns probabilities to possible next tokens, then picks the most likely one
- This is why the output is "The cat sat on the mat" — mat had the highest probability

Why this matters

AI works with numbers, not meanings. It does not 'know' what a word means — it knows which numbers tend to follow which other numbers.

This is also why unusual words or long words can confuse the model — they cost more tokens and have less pattern data behind them.

3. Why LLMs Make Mistakes — Especially in Maths

Because LLMs predict words rather than solve problems, they can get things wrong — and they will say the wrong answer with complete confidence.

Without the right prompt

"What is 234×678 ?"

With a better prompt

The LLM guesses a number from patterns. It may be wrong, but it will not tell you that.

"Solve 234×678 step by step and show your working."

Asking for step-by-step reasoning forces the model to slow down and is far more likely to produce the correct answer.

Remember

LLMs sound confident even when they are wrong. Always verify important facts, calculations, and figures independently.

4. Why Do We Need Prompt Engineering?

The quality of an LLM's output depends almost entirely on the quality of the prompt you give it. A vague question gets a vague answer. A precise, well-structured prompt gets a focused, useful answer.

Here is the same request — written two ways:



The difference is not the topic — it is the specificity. The good prompt told the AI exactly what format to use, what temperature, and what tone. The AI followed those instructions precisely.

Prompt Engineering is the art and science of writing better prompts to get accurate, clear, and useful outputs from AI. Think of it as learning how to communicate effectively with a very powerful but very literal tool.

5. Structure of a Good Prompt — The 5-Part Framework

Every strong prompt is built from five components. You don't always need all five, but the more you include, the better the output.

1	Task	What do you want the AI to do?	<i>Explain / Summarize / Generate / Compare / Write</i>
2	Role	Who should the AI act as?	<i>Act as a Class 10 science teacher</i>
3	Context	What is the background situation?	<i>Audience: Class 8. Topic: photosynthesis. Level: beginner</i>
4	Guidelines	What rules should the output follow?	<i>Simple tone, under 100 words, no jargon</i>
5	Output Format	How should the answer be structured?	<i>Numbered steps, then one real-life example</i>

Putting it all together

Weak prompt:

"Explain photosynthesis."

Strong prompt using all 5 parts:

"Act as a Class 10 biology teacher [Role]. Explain how photosynthesis works [Task] to students who understand basic plant biology [Context]. Use simple language and avoid Latin terms [Guidelines]. Structure your answer as numbered steps followed by one real-life example [Output Format]."

6. Using Delimiters for Clarity

When your prompt has multiple distinct parts — a task, an audience, a format — delimiters help the AI understand where one instruction ends and the next begins.

Common delimiters

- `"""` Triple quotes — to wrap a block of text
- `` ``` Triple backticks — common in technical prompts
- `---` Triple dashes — to separate sections
- `###` Hashtags — to label sections like headings
- `< >` Angle brackets — to wrap specific variables

Example — structured with delimiters

You are an AI tutor.

`### Task`

Explain what Generative AI is.

Audience

Class 12 students with no prior AI knowledge.

Format

3 bullet points. Each bullet under 20 words.

Compare this to just writing: "Explain Generative AI to students". The delimited version leaves no room for misinterpretation.

7. Prompting Techniques

There are three core prompting techniques you need to know. Each is suited to a different situation.

Technique 1: Zero-Shot Prompting

Ask directly. No examples. No hand-holding.

You ask the AI a question without giving any examples. The AI relies entirely on what it learned during training.

When to use it

- Quick factual questions
- General explanations where style and format don't matter much
- First attempts — you can always add structure if the output isn't right

Example

```
"Explain how rain is formed."
```

Works well when

You need a quick answer and the format doesn't matter

Watch out for

Output may be too long, too technical, or pitched at the wrong level

Technique 2: Few-Shot Prompting

Show examples. The AI copies the pattern.

You give the AI 2–3 examples of the output you want before asking your actual question. The AI learns your preferred style, format, and length from those examples.

Simple example

```
Apple → Fruit  
Carrot → Vegetable  
Mango → ?
```

The AI outputs: Fruit — because it identified the pattern from your examples.

Educational example

```
Question: What is AI?
```

Answer: AI is the ability of machines to think and learn like humans.

Question: What is Machine Learning?

Answer: Machine Learning is a part of AI where machines learn from data.

Question: What is Deep Learning?

Answer: ?

Why it works

- The AI copies the style, length, and tone of your examples
- Outputs become more consistent and predictable
- You control the format without writing lengthy instructions

Best used for

Teaching and education

Structured Q&A formats

Anywhere you need consistent, repeatable outputs

Technique 3: Chain-of-Thought Prompting

Make the AI think step by step before answering.

You explicitly ask the AI to reason through a problem step by step instead of jumping to a final answer. This dramatically reduces errors on logic and maths problems.

Example

Without CoT

"What is 24×15 ?"

AI gives a number. May be wrong. No working shown.

With CoT

"Solve 24×15 . Show each step and explain your reasoning."

AI breaks the problem into parts. Accuracy improves significantly.

Also works for non-maths problems

"You are a teacher. Explain step by step how photosynthesis works for Class 10 students. Walk through each stage before summarising."

Important

AI does not naturally reason — it generates text. Chain-of-thought works because asking for steps forces the model to produce reasoning-shaped text, which is more likely to be correct.

8. Quick Summary — Which Technique to Use

Zero-Shot	Few-Shot	Chain-of-Thought
No examples given	2–3 examples given	Step-by-step reasoning
AI uses only its training	AI copies your pattern	AI thinks out loud
Fast, simple	Consistent, structured	Accurate, logical
Good for quick lookups	Good for education	Good for maths & logic

9. Prompt Engineering and Product Management

In AI Product Management, prompts are not just a way to talk to AI — they are a core part of the product itself.

Prompts as UX	Prompts as Logic	Prompts as Features
- Shapes the user's entire experience - A confusing prompt = a frustrated user	- Defines what the AI does behind the scenes - Bad prompt = wrong output = broken product	- Chatbot responses - Resume reviewers - AI tutors - Customer support bots

Bottom line

A bad prompt = a bad product experience. The better you understand how to write prompts, the better the AI products you will be able to build and use.

Key Terms to Remember

Term	Meaning
LLM	Large Language Model — an AI trained on text to predict and generate language
Token	The unit an LLM uses to process text — roughly a word or part of a word
Prompt	The text you type into an AI to get a response
Prompt Engineering	The skill of writing prompts that produce accurate, useful, structured outputs
Zero-Shot	Asking a question directly with no examples

Few-Shot	Giving 2–3 examples so the AI copies the pattern before asking your question
Chain-of-Thought	Asking the AI to reason step by step before giving a final answer
Delimiter	A symbol that separates parts of a prompt clearly (e.g. ###, ``` , ---)